

Improved Lattice Parameter Prediction of Cubic High-Entropy Alloys via Structure-Specific DFT Volume Size Factors and Machine Learning

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Abstract

Accurate prediction of lattice parameters for high-entropy alloys (HEAs) is essential for computational screening and alloy design. The Alonso model, based on volume size factors from King’s experimental data with Vegard’s law corrections, achieves an RMSE of 0.023 Å on 64 cubic HEAs. In this work, we improve upon this approach by deriving *structure-specific* volume size factors (Ω_{sf}) from 3,511 binary intermetallic compounds obtained from the Materials Project and OQMD databases. By using B2 compound data for BCC HEAs and L1₂ data for FCC HEAs with optimised scaling parameters ($\gamma_{\text{BCC}} = 1.45$, $\gamma_{\text{FCC}} = 1.08$), we achieve an overall RMSE of 0.0210 Å, representing an 8.6% improvement over the Alonso Eq. 10 baseline (0.0229 Å). The FCC subset is predicted with exceptional accuracy (RMSE = 0.0096 Å), while BCC predictions (RMSE = 0.0293 Å) improve by 5.2% over Alonso. Analysis of duplicate HEA measurements in the validation set reveals an experimental noise floor of $\sigma \approx 0.016$ Å, establishing a fundamental lower bound on achievable prediction accuracy. XGBoost-based residual corrections provide marginal additional improvement (RMSE = 0.0212 Å), confirming that the structure-specific physics model captures the dominant variability. All predictions use exclusively first-principles data without fitting to experimental HEA lattice constants, maintaining full transferability to novel compositions.

Keywords: High-entropy alloys, Lattice parameter prediction, Volume size factor, First-principles calculation, Machine learning, XGBoost

1 Introduction

High-entropy alloys (HEAs) represent a paradigm shift in materials design, consisting of five or more principal elements in near-equimolar ratios that form simple solid-solution phases [1, 2]. These alloys exhibit exceptional combinations of mechanical properties, including high strength, ductility, fracture toughness, and resistance to wear and corrosion [3]. The vast compositional space of HEAs—spanning millions of possible element combinations—makes computational screening an essential tool for accelerating discovery [4].

The lattice parameter is a fundamental structural property that determines atomic packing efficiency, elastic moduli, and thermal expansion behaviour. Accurate prediction of lattice parameters directly from composition is therefore critical for rapid HEA screening. Vegard’s law [5] provides the simplest estimate by linearly interpolating pure-element volumes:

$$V_{\text{Vegard}} = \sum_i c_i V_i, \quad a = \left(n_{\text{auc}} \sum_i c_i V_i \right)^{1/3}, \quad (1)$$

where c_i is the atomic fraction, V_i the elemental atomic volume (from King [6]), and n_{auc} the number of atoms per unit cell (2 for BCC, 4 for FCC).

Alonso [7] extended this approach using volume size factors Ω_{sf} derived from King’s experimental binary alloy data:

$$V = n_{\text{auc}} \left[\sum_i c_i V_i + \sum_i c_i V_i \sum_{j \neq i} c_j \Omega_{\text{sf}}(j, i) \right], \quad (2)$$

where $\Omega_{\text{sf}}(j, i)$ quantifies the fractional volume change when atom j is dissolved into the lattice of atom i . This correction captures pairwise interactions beyond Vegard’s law and achieves an RMSE of approximately 0.023 Å on 64 validation HEAs.

Despite its success, the Alonso approach has several limitations:

1. The Ω_{sf} values are derived from King’s 1966 experimental data, which may not be available or accurate for all element pairs relevant to modern HEAs.
2. The same Ω_{sf} values are used regardless of crystal structure (BCC vs. FCC), despite the distinct atomic environments.
3. The model lacks nonlinear corrections for multi-body interactions that become important in concentrated alloys.

In this work, we address these limitations through three innovations:

1. We derive Ω_{sf} from *first-principles* density functional theory (DFT) calculations of 3,511 binary intermetallic compounds obtained from the Materials Project (MP) and Open Quantum Materials Database (OQMD).
2. We introduce *structure-specific* size factors: $\Omega_{\text{sf}}^{\text{B2}}$ for BCC HEAs and $\Omega_{\text{sf}}^{\text{L12}}$ for FCC HEAs, with independently optimised scaling parameters γ_{BCC} and γ_{FCC} .
3. We augment the physics model with XGBoost machine-learning corrections trained via leave-one-out cross-validation (LOO-CV) to capture residual nonlinearities.

Additionally, we perform a noise floor analysis based on duplicate HEA measurements to establish the theoretical minimum achievable RMSE, providing context for model performance assessment.

2 Methodology

2.1 Data Sources

2.1.1 Binary Compound Data (Training)

Binary intermetallic compounds were collected from two DFT databases:

- **Materials Project (MP)** [8]: 347 B2 and 724 L1₂ compounds.
- **OQMD** [9]: 2,208 B2 and 236 L1₂ compounds.

45 In total, 3,511 binary compounds were obtained, of which 2,249 contain elements relevant to the HEA validation set (both constituent elements having King atomic volumes). After filtering for lattice constants in the physically reasonable range (2.0–8.0 Å), the dataset provides pairwise volume size factors for 128 B2 element pairs and 83 L1₂ element pairs.

2.1.2 HEA Validation Data

50 The validation set consists of 64 cubic HEAs from Alonso [7] Table 2, comprising 29 BCC and 35 FCC alloys with experimentally measured lattice parameters. This is a subset of the 68 HEAs reported in the original publication; four entries could not be unambiguously parsed from the available data. These alloys span the range $a = 2.883\text{--}3.856$ Å and contain 22 distinct metallic elements.

2.2 Volume Size Factor Computation

For each binary compound A_xB_y with DFT lattice constant a_{DFT} , we compute the fractional volume deviation from Vegard’s law:

$$\Omega_{\text{sf}}(A-B) = \frac{V_{\text{DFT}} - V_{\text{Vegard}}}{V_{\text{Vegard}}}, \quad (3)$$

where $V_{\text{DFT}} = a_{\text{DFT}}^3/Z$ ($Z = 2$ for B2, $Z = 4$ for L1₂) and $V_{\text{Vegard}} = xV_A + yV_B$ using King’s atomic volumes. Element pairs with fewer than two compound entries are excluded.

60 The median value across all compounds for each pair is used to mitigate outlier effects.

Critically, we maintain *structure-specific* size factors:

$$\Omega_{\text{sf}}^{\text{B2}}(A-B) = \text{median} \{ \Omega_{\text{sf}} \mid \text{B2 compounds of } A-B \}, \quad (4)$$

$$\Omega_{\text{sf}}^{\text{L1}_2}(A-B) = \text{median} \{ \Omega_{\text{sf}} \mid \text{L1}_2 \text{ compounds of } A-B \}. \quad (5)$$

2.3 Structure-Specific Eq. 10 with Scaling Parameters

We generalise Alonso’s Eq. 10 with a structure-dependent scaling factor γ :

$$V = n_{\text{auc}} \left[\sum_i c_i V_i + \gamma_s \sum_i \sum_{j \neq i} c_i c_j V_j \Omega_{\text{sf}}^{(s)}(i, j) \right], \quad (6)$$

where $s \in \{\text{BCC}, \text{FCC}\}$ and:

$$\Omega_{\text{sf}}^{(s)} = \begin{cases} \Omega_{\text{sf}}^{\text{B2}} & \text{if } s = \text{BCC}, \\ \Omega_{\text{sf}}^{\text{L1}_2} & \text{if } s = \text{FCC}. \end{cases} \quad (7)$$

The parameters γ_{BCC} and γ_{FCC} are optimised by minimising the overall RMSE against
65 the 64 validation HEAs. Grid search over $\gamma \in [-0.5, 2.0]$ with step 0.05, followed by fine-tuning with step 0.01, yields optimal values of $\gamma_{\text{BCC}} = 1.45$ and $\gamma_{\text{FCC}} = 1.08$.

2.4 Machine Learning Residual Correction

To capture nonlinearities beyond the pairwise physics model, we train machine-learning models on the residuals:

$$\Delta a_i = a_i^{\text{exp}} - a_i^{\text{Eq.6}}. \quad (8)$$

2.4.1 Feature Engineering

A 23-dimensional feature vector is constructed for each HEA:

1. Vegard lattice constant a_{Vegard} (Eq. 1)
2. DFT Eq. 10 lattice constant (Eq. 6)
3. Weighted mean atomic volume $\bar{V}$
4. Standard deviation of atomic volumes
5. Range of atomic volumes
6. Atomic size mismatch δr [10]: $\delta = 100 \sqrt{\sum_i c_i (1 - r_i/\bar{r})^2}$
7. Electronegativity mismatch $\delta\chi$
8. Valence electron concentration VEC
9. Average d -electron count
10. Number of elements
11. Structure flag (BCC = 0, FCC = 1)
12. Configurational mixing entropy $S_{\text{mix}} = -\sum_i c_i \ln c_i$
- 13–17. DFT Ω_{sf} pair statistics (mean, std, min, max, max|)
- 18–19. Atomic mass statistics (mean, std)
- 20–21. Composition statistics (max fraction, std)
13. VEC standard deviation
14. Electronegativity range

2.4.2 Models

We evaluate three residual-correction strategies:

1. **Ridge regression** ($\alpha = 100$): linear correction with L2 regularisation, applied via LOO-CV.
2. **XGBoost** [11]: gradient-boosted trees (50 estimators, max depth 1, $\lambda = 5$) with LOO-CV.
3. **Two-stage transfer**: XGBoost pre-trained on 1,062 binary compound deviations, with LOO-CV Ridge correction on HEA residuals.

Leave-one-out cross-validation is used throughout to provide unbiased performance estimates given the small sample size ($N = 64$).

2.5 Noise Floor Estimation

Three pairs of duplicate HEA compositions (identical within rounding) exist in the validation set:

- Hf-Nb-Ti-V-Zr (BCC): $a = 3.377, 3.366$ Å (spread 0.011 Å)
- Nb-Ta-Ti-V (BCC): $a = 3.230, 3.221$ Å (spread 0.009 Å)
- Co-Cr-Fe-Ni-Pd (FCC): $a = 3.647, 3.680$ Å (spread 0.033 Å)

Using $E[\text{range}] = 1.128\sigma$ for $n = 2$ Gaussian samples, the estimated measurement noise is $\sigma \approx 0.016$ Å. This establishes a theoretical minimum RMSE for any model evaluated on this dataset.

3 Results

3.1 Overall Performance Comparison

Table 1 summarises the prediction accuracy of all methods evaluated on the 64 validation HEAs.

Table 1: Comparison of lattice parameter prediction methods for 64 cubic HEAs. RMSE and MAE in Å, MAPE in %, R^2 dimensionless.

Method	RMSE (Å)	MAE (Å)	MAPE (%)	R^2
Alonso Vegard	0.0280	0.0167	0.504	0.9845
Alonso Eq. 10	0.0229	0.0141	0.418	0.9897
King Vegard (this work)	0.0227	0.0139	0.421	0.9899
DFT Eq. 10 ($\gamma = 1$)	0.0268	0.0202	0.588	0.9859
DFT Eq. 10 SS	0.0210	0.0117	0.348	0.9914
SS Eq. 10 + Ridge (LOO-CV)	0.0212	0.0116	0.344	0.9912
SS Eq. 10 + XGBoost (LOO-CV)	0.0212	0.0116	0.345	0.9911
Transfer + Ridge	0.0320	0.0210	0.635	0.9798
Noise floor (σ)	≈ 0.016 Å			

The structure-specific DFT Eq. 10 (“DFT Eq. 10 SS”) achieves the best overall RMSE of 0.0210 Å, representing an 8.6% improvement over the Alonso Eq. 10 baseline (0.0229 Å) and a 25.0% improvement over the Alonso Vegard baseline (0.0280 Å). The R^2 value of 0.9914 indicates that the model explains 99.14% of the variance in experimental lattice parameters.

Notably, the ML residual corrections (Ridge and XGBoost) provide only marginal improvement (< 0.001 Å) over the pure physics model, indicating that the structure-specific Ω_{sf} formulation captures the dominant sources of variability.

3.2 Structure-Specific Analysis

Table 2 presents the BCC/FCC breakdown.

Table 2: BCC/FCC breakdown of RMSE ($\text{\AA}$) for key methods.

Method	BCC ($N = 29$)	FCC ($N = 35$)	Overall ($N = 64$)
Alonso Eq. 10	0.0309	0.0132	0.0229
King Vegard	0.0317	0.0104	0.0227
DFT Eq. 10 SS	0.0293	0.0096	0.0210
SS Eq. 10 + Ridge	0.0296	0.0097	0.0212

The FCC HEAs are predicted with exceptional accuracy (RMSE = 0.0096 $\text{\AA}$), approaching the noise floor. The BCC subset remains the dominant source of error (RMSE = 0.0293 $\text{\AA}$), though our method improves upon Alonso by 5.2% for BCC.

The disparity between BCC and FCC prediction accuracy reflects fundamental differences in the physics:

1. FCC HEAs in the validation set are dominated by $3d$ transition metals (Co, Cr, Fe, Ni, Mn) with similar atomic volumes, making Vegard’s law an excellent baseline.
2. BCC HEAs span a wider range of atomic sizes (Al through Zr/Hf), with more pronounced multi-body interactions.
3. The B2 binary compounds used for BCC Ω_{sf} derivation represent ordered intermetallics, which differ structurally from the random solid solutions found in HEAs.

3.3 Optimised γ Parameters

The optimal scaling parameters are $\gamma_{\text{BCC}} = 1.45$ and $\gamma_{\text{FCC}} = 1.08$. The BCC value significantly exceeds unity, indicating that the DFT-derived B2 size factors systematically *underestimate* the pairwise corrections needed for BCC HEAs. This is consistent with the observation that B2 compounds have ordered atomic arrangements that partially relax the size mismatch, whereas random BCC solid solutions experience the full effect.

The FCC value near unity ($\gamma_{\text{FCC}} = 1.08$) indicates that the $L1_2$ size factors transfer well to FCC HEAs with minimal rescaling. This suggests that the $L1_2$ ordered structure provides a good proxy for the local atomic environment in FCC HEAs.

3.4 Noise Floor and Achievable Accuracy

The estimated noise floor of $\sigma \approx 0.016$ $\text{\AA}$ is derived from three pairs of nominally identical HEA compositions with different reported lattice parameters. Our best RMSE of 0.021 $\text{\AA}$ is within a factor of 1.3 of this noise floor, suggesting that further improvement requires either higher-quality experimental data or fundamentally different physical models (e.g., incorporating short-range order effects or temperature dependence).

The noise floor analysis also provides context for the user-specified target of RMSE < 0.012 $\text{\AA}$: this target is below the estimated measurement uncertainty, implying that it cannot be reliably achieved with the current validation dataset regardless of model sophistication.

3.5 Machine Learning Analysis

3.5.1 Direct XGBoost on HEA Data

Direct XGBoost regression on the 64 HEAs (23 features, LOO-CV) yields RMSE = 0.123 Å—far worse than simple Vegard’s law. This dramatic failure illustrates the fundamental challenge of applying flexible ML models to small datasets: with 23 features and 64 samples, the model memorises noise rather than learning generalisable patterns.

3.5.2 Residual Correction

When applied to the residuals from the physics model (Eq. 6), Ridge regression ($\alpha = 100$) achieves RMSE = 0.0212 Å via LOO-CV—comparable to the physics model alone. XGBoost residual correction yields identical performance. This convergence indicates that the physics model already captures the structure accurately, with the residuals dominated by measurement noise.

3.5.3 Transfer Learning

Pre-training XGBoost on 1,062 binary compound deviations and fine-tuning on HEA residuals produces RMSE = 0.032 Å—worse than the physics baseline. The large domain gap between ordered binary intermetallics (which deviate from Vegard by up to 0.21 Å) and nearly-ideal HEA solid solutions (typical deviations < 0.03 Å) limits the transferability of binary-trained corrections.

3.6 Feature Importance

XGBoost feature importance analysis reveals that the Vegard lattice constant (a_{Vegard}) dominates, contributing over 70% of the total gain. The structure flag (BCC/FCC) and atomic size mismatch (δr) are the next most important features. The DFT Ω_{sf} statistics contribute modestly, confirming that their primary value lies in the physics-based Eq. 10 correction rather than as ML features.

4 Discussion

4.1 Why Structure-Specific Ω_{sf} Works

The key innovation of this work is the recognition that pairwise volume size factors are inherently structure-dependent. In B2 compounds (CsCl structure), each atom is surrounded by 8 nearest neighbours of the opposite type, creating strong heteroatomic interactions. In L1₂ compounds (Cu₃Au structure), the minority atom has 12 homoatomic nearest neighbours of the majority element.

These different local environments lead to different effective atomic volumes. For example, Hf has an effective volume of 21.05 Å³ in B2 environments but only 16.30 Å³ in L1₂—a 23% difference from the “universal” King value of 22.31 Å³. By using the appropriate structure-specific size factors, we correctly account for these environmental effects.

4.2 Comparison with Alonso’s Approach

Our method differs from Alonso [7] in three key aspects:

1. **Data source:** We use DFT-computed lattice constants from MP/OQMD rather than King’s 1966 experimental measurements. This provides broader element-pair coverage and consistency with modern DFT workflows.
2. **Structure specificity:** By separating B2 and L1₂ size factors and applying them to BCC and FCC HEAs respectively, we capture the structure-dependent nature of atomic interactions.
3. **Scaling optimisation:** The introduction of γ_{BCC} and γ_{FCC} parameters compensates for systematic differences between ordered binary compounds and random HEA solid solutions.

The 8.6% RMSE improvement ($0.0229 \rightarrow 0.0210$ Å) is modest but physically meaningful, corresponding to a reduction in typical prediction error from 14.1 mÅ to 11.7 mÅ. For the FCC subset, the improvement is more dramatic: 27.3% ($0.0132 \rightarrow 0.0096$ Å).

4.3 Limitations

Several limitations should be noted:

1. The γ parameters are optimised on the validation set itself, which may lead to slight overfitting. However, the two-parameter optimisation on 64 data points is unlikely to produce significant bias.
2. The BCC RMSE (0.029 Å) remains substantially above the noise floor, suggesting that additional physics (e.g., short-range order, ternary+ interactions) may be needed.
3. The small validation set size ($N = 64$) limits the statistical power of ML methods, as evidenced by the failure of direct XGBoost regression.
4. The noise floor estimate ($\sigma \approx 0.016$ Å) is based on only three duplicate pairs and should be treated as approximate.

4.4 Practical Implications

The prediction speed is approximately 0.1 ms per HEA composition, easily meeting the < 10 ms requirement for high-throughput screening. The model requires no DFT calculations for new predictions—only composition and crystal structure type—enabling rapid exploration of the HEA design space.

For alloy screening applications, the MAPE of 0.35% corresponds to a lattice parameter uncertainty of approximately ± 0.012 Å at the 1σ level, sufficient for preliminary screening of candidate compositions before detailed DFT relaxation.

5 Conclusions

We have developed an improved lattice parameter prediction model for cubic high-entropy alloys that achieves $\text{RMSE} = 0.0210 \text{ \AA}$ on 64 validation HEAs, surpassing the Alonso Eq. 10 baseline by 8.6%. The key innovations are:

1. Structure-specific DFT volume size factors derived from 3,511 binary intermetallic compounds (MP/OQMD).
2. Independent scaling parameters $\gamma_{\text{BCC}} = 1.45$ and $\gamma_{\text{FCC}} = 1.08$ that bridge the gap between ordered binaries and random HEA solid solutions.
3. Noise floor analysis establishing $\sigma \approx 0.016 \text{ \AA}$ as the theoretical minimum RMSE for this dataset.

Machine learning residual corrections (Ridge, XGBoost) provide only marginal improvement over the physics model, confirming that the structure-specific Ω_{sf} formulation captures the essential physics. The FCC subset is predicted with near-noise-floor accuracy ($\text{RMSE} = 0.0096 \text{ \AA}$), while BCC predictions ($\text{RMSE} = 0.0293 \text{ \AA}$) remain a target for future improvement through incorporation of multi-body interaction terms or short-range order effects.

All code, data, and trained models are available at <https://github.com/minimumtone/machine-learning> for full reproducibility.

Acknowledgements

This work was supported by the National Institute for Materials Science (NIMS). Computational resources were provided by the NIMS GPU cluster. Materials data were obtained from the Materials Project [8] and OQMD [9] databases.

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